

Day 9 — Mixed Practice

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set comparing complement vs traditional borrow.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam Navataścaramaṁ Daśataḥ* (IAST: Nikhilam Navataścaramaṁ Daśataḥ; lit. “all from 9, the last from 10”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed problem set.** 20 subtractions, classes mixed:
 - Simple borrows (use borrow or Nikhilam, your choice)
 - Multi-borrow (Nikhilam likely faster)
 - Subtraction from powers of 10 (Nikhilam dominates)
 - Add-the-complement (general)
2. **Strategy choice.** Each student picks the method per problem and writes the chosen method beside the answer.
3. **Cross-check.** Pairs swap; compare method choices. “*Did you both pick the same method? When did you disagree?*”
4. **Discussion.** “*Is Nikhilam ALWAYS better? When is plain borrowing fine?*” (Single-digit, small subtractions.)
5. **Method-comparison artefact.** Each student starts a “personal mental-math toolkit” page summarising when to use each method.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).